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8 A transformer steps down high-voltage electricity. The transformer has 6200 turns on the primary coil and 43 turns on the secondary coil. The output voltage is 230 V.

(a) (i) Calculate the input voltage in the step-down transformer.

input voltage = ... V [2]

(ii) A 2.2 kW kettle is connected to the secondary coil of the transformer.

Calculate the current in the primary coil of the step-down transformer. Assume the transformer is 100% efficient.

current = ... A [3]

(b) A power station produces 1.3 GW of electricity. A transformer at the power station steps up the voltage before the electricity is transmitted. This reduces the power loss during transmission.

(i) State the equation for power loss in electricity transmission.

..... [1]

(ii) Explain how the step-up transformer reduces power losses during electricity transmission.

.....

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..... [2]

[Total: 8]

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